Benchmarking Foundation Models using Satellite Image Time Series for Land Monitoring - MVA 2025/2026

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1 Introduction

Accurate and up-to-date mapping of land surfaces is a critical challenge for monitoring and mitigating the effects of climate change. Over the past decade, satellites have provided high-resolution Satellite Time Series (SITS) of multi-spectral imagery. While Deep Neural Networks (DNNs) have emerged as powerful tools to exploit this massive volume of data, their performance remains heavily dependent on the availability of large-scale labeled datasets.

In remote sensing, the **scarcity of labeled data is a primary bottleneck**. Launching and maintaining satellite constellations is costly, and revisit frequencies may not always capture the necessary temporal granularity for every zone. Furthermore, generating ground-truth labels is a labor-intensive process requiring extensive field surveys. Beyond data scarcity, many existing models are designed for a single specific task and struggle to integrate the diverse modalities provided by various sensors to enrich image representations.

To address these limitations, there is a burgeoning interest in **Foundation Models (FMs)**. These models are designed to process data from heterogeneous sensors and learn robust representations that can be adapted to a wide range of downstream tasks, even those suffering from label scarcity. FMs aim to standardize SITS processing by generating rich embeddings. For every pixel, the model computes a high-dimensional vector that encapsulates spatial, temporal, and physical characteristics. **They lack a standard benchmark to compare their performance.**

In this project, we propose a benchmark to evaluate the performance of several FMs on the task of crop segmentation. Specifically, we test **ALISE, TESSERA**, and **AlphaEarth** against U-TAE, a state-of-the-art specialized architecture that does not follow the foundation model paradigm.

2 Benchmarked models

In this section, we present a curated overview of the models evaluated in this benchmark.

2.1 ALISE

ALISE (ALigned SITS Encoder) [1] is a foundation model designed to produce consistent representations for **multi-year, irregular, and unaligned SITS**.

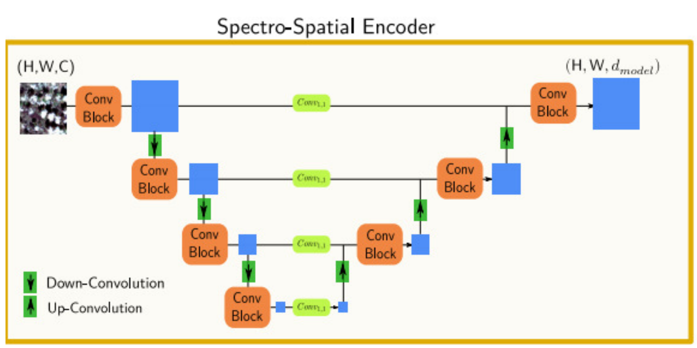


Fig. 1 U-BARN Architecture. Courtesy of [1]

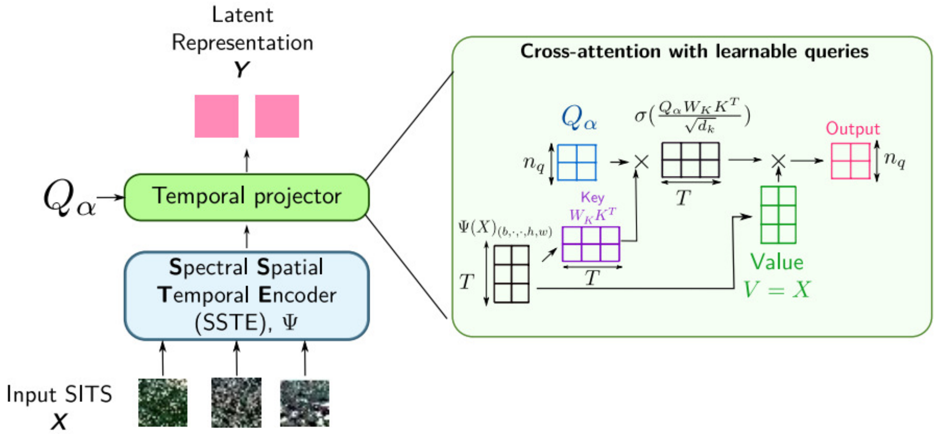


Fig. 2 ALISE Architecture. Courtesy of [1]

The architecture is composed of two main blocks:

1. A **Spatial, Spectral, and Temporal Encoder (SSTE)**, which follows the **U-BARN** architecture (Fig. 1),
2. A **Temporal Projector** that processes the irregular and unaligned outputs of the SSTE to generate aligned SITS representations (Fig. 2).

U-BARN combines a **U-Net structure with a Transformer block** [2] to capture multi-scale features. The U-Net, implemented with four down-sampling and up-sampling levels, extracts high-level spatial features. Input time series of shape (B, T, C, H, W) are reshaped into $(B \times T, C, H, W)$ tensors for U-Net processing. To incorporate temporal information, the Day of the Year (DoY) is embedded into the feature space. After U-Net processing, the tensors of dimension $(B \times T, d_{model}, H, W)$ are reshaped to $(B \times H \times W, T, d_{model})$ before entering the Transformer block. This block processes all observations via a self-attention mechanism, enabling it to capture complex phenological patterns over time. The Transformer is pre-trained using Masked Time Series Modeling, learning to reconstruct the SITS even when several dates are corrupted (e.g., by cloud cover).

U-BARN outputs still depend of the number of image in the SITS. The role of the Temporal Projector is to align these SITS with different sizes. The projection is performed by a **multihead temporal cross-attention mechanism between learnable queries and the output of the SSTE**. Learning queries are vectors optimized during the training, they represent latent temporal features. This process allows the model to selectively aggregate the most relevant temporal information, regardless of the acquisition frequency or the presence of clouds. The outputs from all attention heads are then concatenated along the feature dimension and processed by a Multi-Layer Perceptron (MLP) to produce the final aligned embedding.

This model is open, with the code, weights available on the web.

2.2 TESSERA

TESSERA [3] (Temporal Embeddings of Surface Spectra for Earth Representation and Analysis) operates under the "Embedding-as-Data" paradigm, providing analysis-ready, pre-computed embeddings for each pixel at a global scale. The model exclusively processes time series at the pixel level and does not utilize explicit spatial features or spatial convolutions. Consequently, it achieves a direct multi-temporal fusion of the data, relying purely on the natural spatial correlation of real-world observations to form spatially coherent outputs.

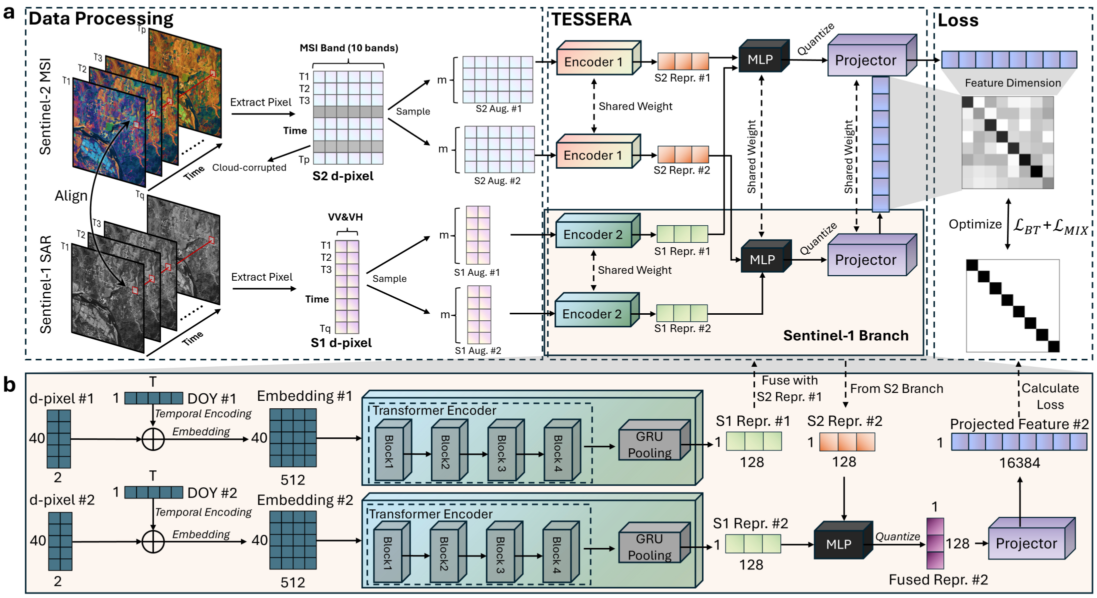


Fig. 3 (a) Overall architecture: multi-temporal Sentinel-1/2 observations are converted into modality-specific d-pixels, augmented twice, and encoded by dual branches with shared weights to produce compact 128-D embeddings. (b) Zoom-in on the Sentinel-1 branch: temporal sequences are processed by a 4-block Transformer followed by GRU pooling to capture dynamic backscatter patterns before fusion into the joint TESSERA embedding. Courtesy of [3].

As illustrated in the overview of the TESSERA processing pipeline (Fig. 3), the architecture handles multi-modal data streams by extracting temporal sequences of optical (Sentinel-2) and radar (Sentinel-1) observations into modality-specific "d-pixels". The training strategy is akin to contrastive learning, specifically employing the Barlow Twins objective function. The model aims to maximize the similarity between two different "views" of the same geographic location. In this framework, a view corresponds simply to a time series of multiple modalities randomly sampled from the available observations within the same year. This sparse temporal sampling strategy forces the model to learn representations that remain robust and invariant to the irregularity and cloud obstructions typical of Earth Observation data. It produces 128-dimensional embeddings for each pixel. This model is open, with the code, weights, and pre-computed embeddings available on the web.

2.3 AlphaEarth

In contrast to TESSERA's pixel-wise and primarily dual-modality focus, AlphaEarth [4] integrates a broader array of sensors and modalities. Its input sources and reconstructed outputs go far beyond standard optical and radar satellite imagery (Sentinel-1, Sentinel-2, Landsat 8/9), incorporating digital elevation models, LiDAR profiles (GEDI), and even continuous climate variables (ERA5-Land). By fusing these irregular and unaligned sources, AlphaEarth creates a different embedding space, representing the Earth's surface as a regular 64-dimensional embedding field.

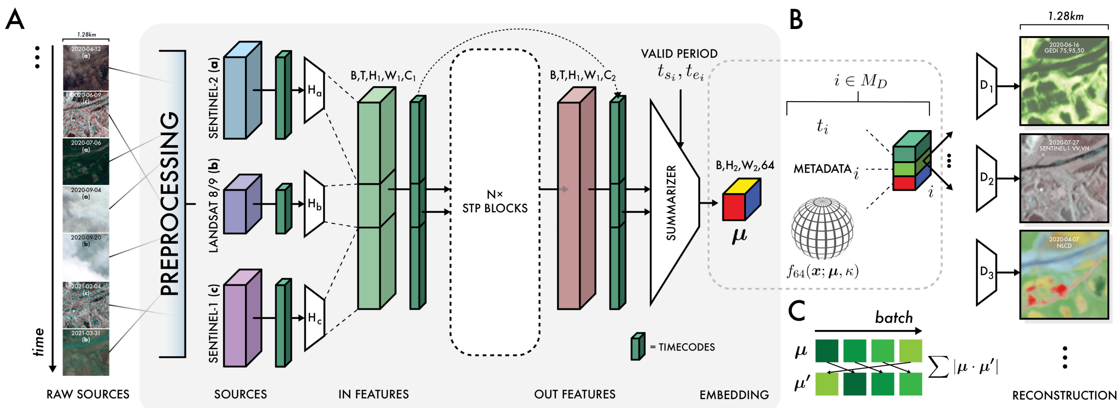


Fig. 4 (A) Block diagram of the overall network architecture used for video analysis. Preprocessing converts raw observation data via normalization using global statistics, and acquisition timestamps are converted to sinusoidal timecodes. Individual source encoders transform inputs to the same latent space before entering the bulk of the model. Outputs are summarized using conditional timecodes or "summary periods", unique to each decoded source and contrastive learning task. μ refers to the embedding outputs of the model. The grey shaded region constitutes the model. (B) Model outputs are treated as the mean direction of a von Mises-Fisher distribution, and decoding proceeds by sampling this distribution, and concatenating it with sensor geometry metadata and a timecode indicating the relative position in the valid period to decode. Decoding proceeds for all sources, with losses dependent on the characteristics of each source (see supplemental materials S1). Courtesy of [4].

The block diagram in Fig. 4 depicts how the architecture summarizes these diverse spatial and temporal contexts into a highly compact 64-byte representation. Furthermore, its learning paradigm differs significantly; rather than a contrastive approach, AlphaEarth relies on a generative learning approach. The network employs an implicit decoder that attempts to reconstruct the original, multi-source observations and metadata directly from this small embedding. By working at the patch level rather than the pixel level and focusing on reconstruction, this generative approach allows AlphaEarth to explicitly capture spatial coherence and physical measurement contexts during the learning phase.

In contrast to TESSERA, AlphaEarth's code and weights are not publicly available, and the embeddings are only available under a commercial API.

2.4 U-TAE

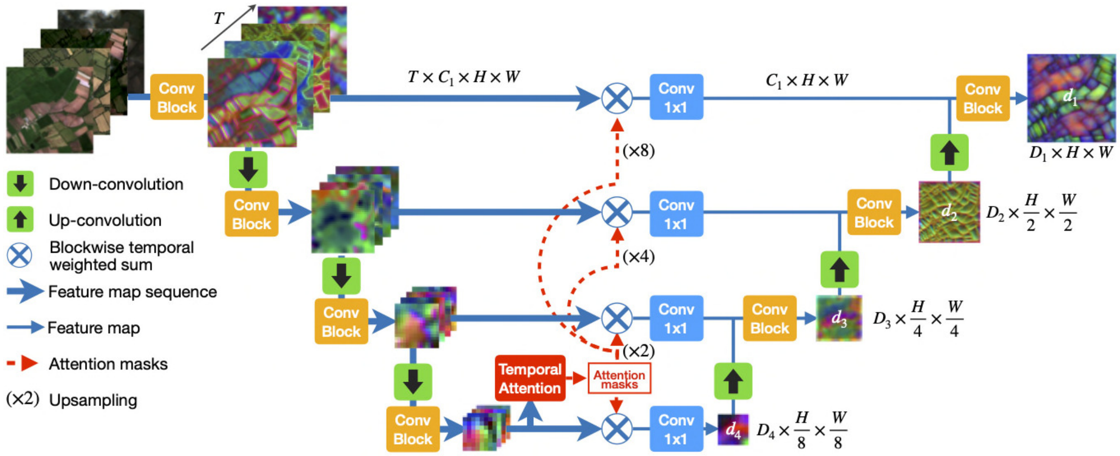


Fig. 5 U-TAE Architecture

Unlike the foundation models previously discussed, **U-TAE (U-Net with Temporal Attention Encoder)** [5] is a supervised architecture specifically designed for SITS segmentation and serves as the primary benchmark for the PASTIS dataset.

The architecture follows a modified U-Net structure (Fig. 5):

1. **Spatial Encoding:** Each image in the sequence is processed independently by a shared multi-level convolutional encoder. For each level l, the encoder produces a feature map of shape $(T \times C_l \times H_l \times W_l)$, where H_l, W_l decrease as the depth l increases.
2. **L-TAE (Light Temporal Attention Encoder):** To manage computational complexity, the temporal attention masks are only computed at the bottleneck (the lowest resolution level L). These masks consist of pixel-wise attention weights that quantify the importance of each acquisition date $t \in [1, T]$ for G different groups of channels These G groups of masks are then spatially interpolated (resized) to match the dimensions of all higher-level encoder maps ($l < L$). This allows the model to apply temporal pooling at every scale without the overhead of re-computing attention at high resolutions.
3. **Decoding:** For each level, the feature sequence is collapsed into a single representation by a weighted temporal sum using the attention masks. Finally, these synthesized maps are processed by a classic U-Net decoder to produce the final segmentation map.

2.5 Models overviews

Table 1 summarizes the main characteristics of the benchmarked models.

Table 1 Comparison of Foundation Models Architectures and pre-training strategies.

Attribute	TESSERA	AlphaEarth	ALISE	U-TAE*
Input Modalities	SAR (S1), Optical (S2)	S1, S2, Thermal (L8/9)	Optical (S2)	Optical (S2)
SSL Task	Contrastive	Hybrid	Hybrid	N/A
Approach	Discriminative	Generative	Generative	Supervised
Spatial training	*No*	*Yes*	*Yes*	*Yes*
Embedding dim. size	128	64	640	–
Number of parameters	∼1.4B (45.7M encoder)	∼480M (1B variant)	∼1.1M (12.2K frozen)	∼1.1M
Code availability	✓	×	✓	✓

*U-TAE is used as a supervised baseline and does not follow the FM paradigm.

3 Experiments and results

3.1 Dataset

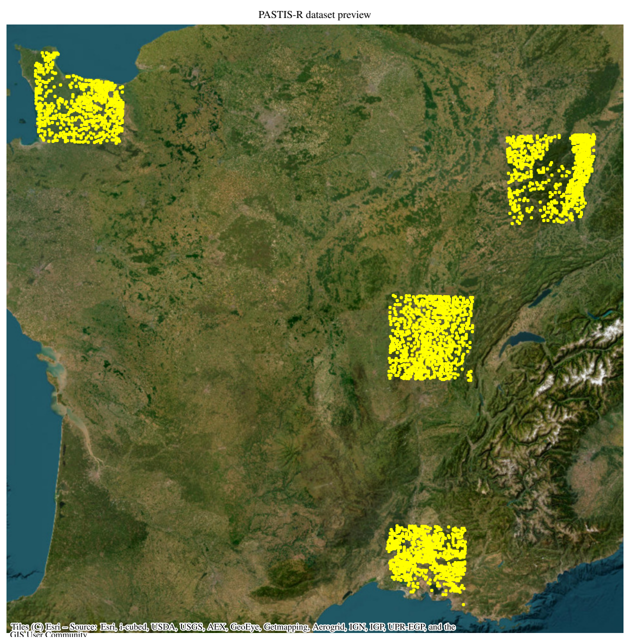


Fig. 6 Map of the PASTIS patches

We utilize the **PASTIS** (Panoptic Agricultural Satellite TIme Series) dataset [5], which consists of 2,433 multi-spectral sequences. Each sequence is structured as a $10 \times 128 \times 128$ tensor, containing between 38 and 61 temporal observations. These data are sourced from the **THEIA** platform, providing Sentinel-2 satellite imagery.

The 10 spectral channels correspond to the non-atmospheric bands of Sentinel-2, following atmospheric correction and spatial resampling at a 10-meter resolution. The observations span four French regions with distinct climatic conditions: *Normandie, Bourgogne, Alsace,* and *Provence* (Fig. 6). Notably, approximately 30% of the images in the dataset are affected by cloud cover, posing a significant challenge for temporal analysis.

Based on the French Land Parcel Information System (RPG), the agricultural fields are categorized into **18 distinct crop classes** (Fig. 7). In addition to these, two specific labels are defined:

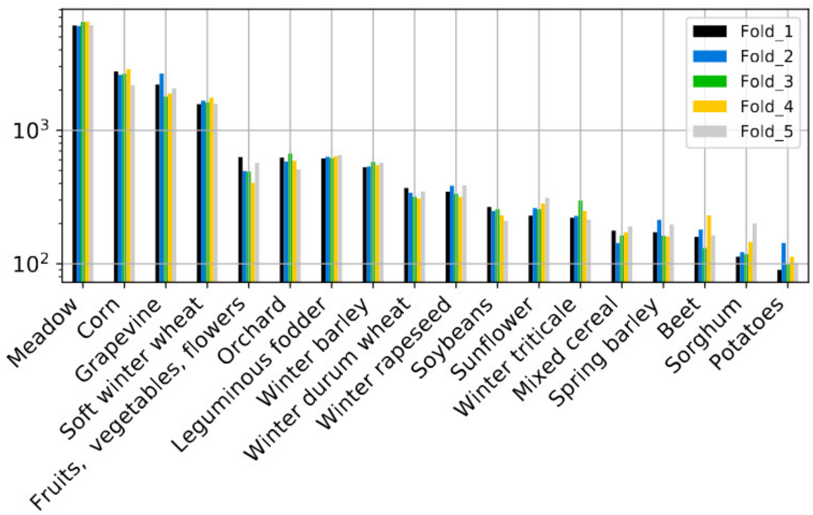


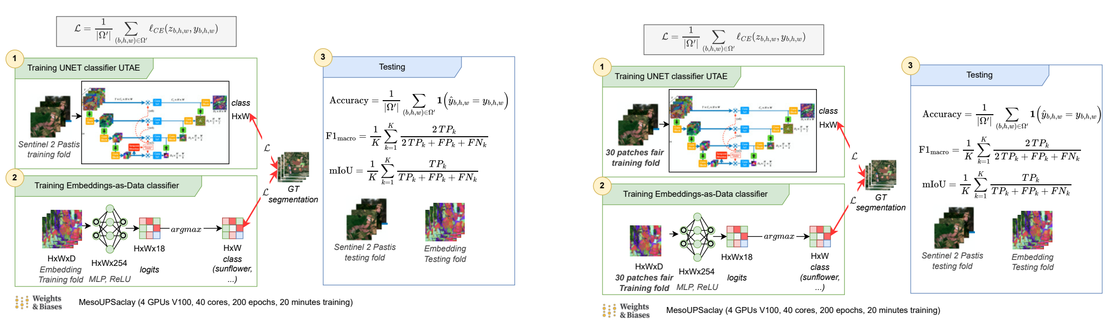
Fig. 7 Distribution of crop growth stages (or classes) within the PASTIS dataset.

- **Void label:** Assigned to pixels belonging to parcels outside the 18 selected classes, or to pixels where the parcel's spatial extension within the patch is less than 50%.
- **Background class:** Refers to pixels that do not belong to any parcel declared in the RPG.

For robust evaluation, the patches are randomly subdivided into **five folds** to facilitate a cross-validation process. Note that the background class and the void label are not utilized during the training.

3.2 Benchmarking setup

Our main benchmarking task is to predict the crop class of each pixel over the PASTIS dataset ("semantic segmentation").



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(a) With the full PASTIS dataset **(b)** With scarce data (30 patches per fold)

Fig. 8 Training overview of our benchmarking framework. Fig. 8a shows the training setup with the full PASTIS dataset, while Fig. 8b shows the training setup with scarce data (30 patches per fold).

Our project mentor gave us the embeddings of the three foundation models (TESSERA, Alpha Earth and ALISE) on the PASTIS dataset. Producing those embeddings is computationally expensive (or even cost money, e.g. for Alpha Earth), this significantly eased our workload. Those embeddings are tensors of shape (PARCEL_ID, H, W, D) where D is the dimension of the embedding. It means that we have one embedding per pixel, which is a vector of dimension D.

As seen in Fig. 8a, we train simple MLPs with `ReLU` activations on top of the FM embeddings (one per FM embedding set, e.g one MLP for TESSERA, one MLP for Alpha Earth, one for ALISE). Those MLPs are optimized using a cross-entropy loss, which is the standard loss for multi-class classification problems. The output of the MLPs is a vector of dimension 18, which corresponds to the 18 crop labels logits. We apply a `argmax` function to the output of the MLP to obtain the final class. Note that the complexity of the classifier does not matter, as we want to evaluate the quality of the embeddings produced by the foundation models, not the capacity of the classifier.

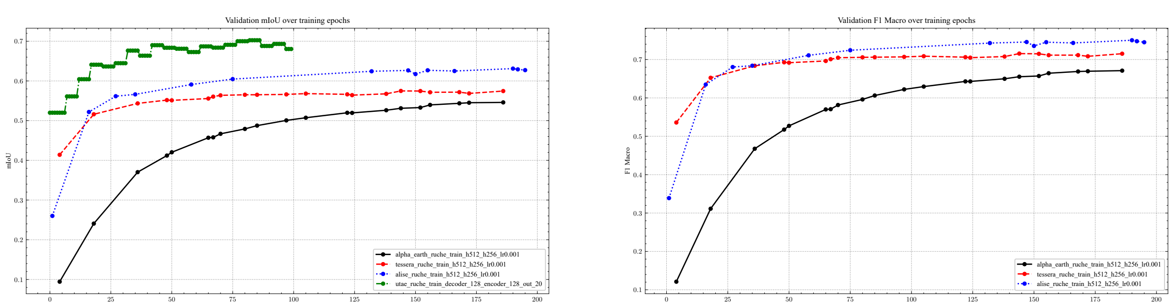
We use the same folds as the PASTIS paper, 3 for training, 1 for validation and 1 for testing. We train the MLPs for 200 epochs with a batch size of 32 (16 for ALISE due to heavier embeddings) and a learning rate of 10^{-3} with Adam optimizer. We used the Ruche cluster of Université Paris Saclay. Specifically the slurm partition `GPU-test` of Ruche cluster, with a training of roughly 20 minutes with four V100 GPUs using `DistributedDataParallel` strategy of [6]. The code is implemented using [7] for industry-grade training and logging, and monitored using [8]. All our training configurations are available in the code repository. We also provide a reproducible notebook to plot all the figures of this report.

To compare with a supervised baseline, we also train a U-TAE model on the same task. We reuse the code of the original paper for training and evaluating. We also use the same hyperparameters. Training takes roughly 14 hours on one V100 GPU. Instead of FM embeddings, it is directly trained on the optical channels of the Sentinel-2 satellite.

We evaluate the performance of the models using three metrics: mean Intersection over Union (mIoU), F1 Macro and Overall accuracy. Note that the `background` and `void` classes of the PASTIS dataset are ignored in the evaluation (as well in the training loss of the MLPs and U-TAE).

We also evaluate the performance of the models in a scarce data setting, where we only use 30 patches per fold for training (configurations suffixed with "_scarce"), see Fig. 8b. Those folds are randomly sampled from the original training folds but maintain a balanced class distribution. This scarce data setting is particularly relevant for satellite image segmentation, as it is often the case that we have limited labeled data.

3.3 Analysis of results



(a) **(b)**

Fig. 9 Increasing trends of the validation F1 Macro (Fig. 9b) and mIoU (Fig. 9a) metrics during training (validation fold). As shows Fig. 9a, U-TAE is trained only for 100 epochs. Fig. 9b does not show the U-TAE curve as the original implementation does not log the F1 Macro metric during training.

Over training epochs, benchmarked models successfully learn to segment the images, as shown by the increasing trends of the validation F1 Macro and mIoU metrics in Fig. 9.

Table 2 Test metrics of the FM-EMBEDDINGS+MLP / U-TAE models on the PASTIS dataset, with the full training set (Table 2a) and with scarce data (Table 2b). Bold values indicate the best performance for each metric.

Model	mIoU	F1 Macro	Overall accuracy
AlphaEarth	0.5287	0.6526	0.8417
Tessera	0.5607	0.7018	0.8257
ALISE	0.6009	**0.7235**	0.8651
U-TAE	**0.7219**	-	**0.9170**

(a) Test metrics

Model	mIoU	F1 Macro	Overall accuracy
AlphaEarth	0.2867	0.3755	0.7267
Tessera	0.4387	**0.5765**	0.7621
ALISE	**0.4482**	0.5729	**0.7898**
U-TAE	0.3869	0.5115	0.7043

(b) Test metrics with scarce data (30 patches per fold)

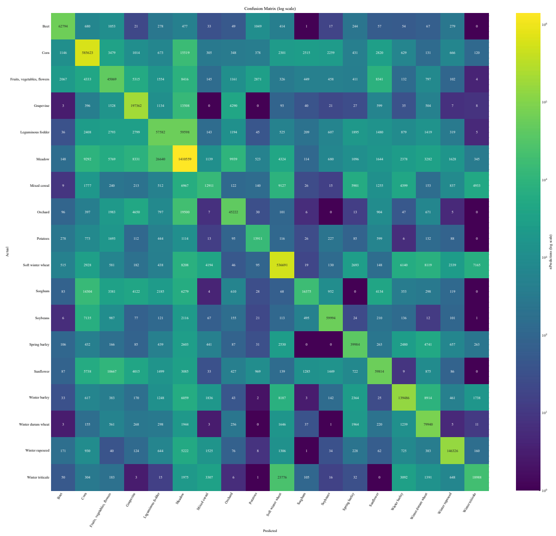
As shown in Table 2, the supervised approach U-TAE achieves the best performance on the full training set, while MLP+ALISE-EMBEDDINGS performs best in the scarce data setting. Despite Tessera (released in November 2025) and ALISE (released in September 2024) being separated by one year of research, ALISE is still a State-Of-The-Art FM in our benchmark. The scarce data setting shows that foundation model achieve strong performance when we have limited labeled data, as they can leverage their pre-training on large amounts of unlabeled data to produce useful embeddings for downstream tasks. Note that there is a net computational cost to using foundation models, as the training of the MLPs on top of the FM embeddings is much faster than training a supervised model like U-TAE from scratch, with less parameters.

As noted in [5, 9], the PASTIS dataset is highly imbalanced, with a ratio of over 50 between the most and least common classes (see Fig. 7). Thus, we provide in Table 3 the F1 score of each class for the different models, which gives a more detailed view of the performance of the models on each class.

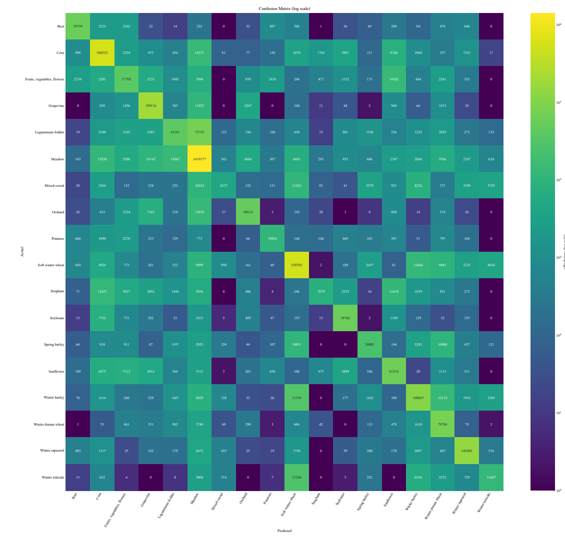
We also show on Fig. 10 the confusion matrix of each FM-EMBEDDINGS+MLP model. Unsurprisingly, confusions seem to occur between semantically close classes such as different cereal types, or Sunflower and Fruits, Vegetable, Flower.

Table 3 Test F1 per class

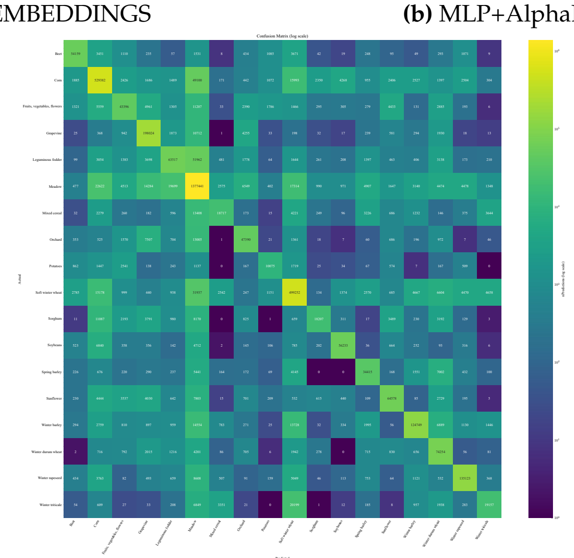
	test_alpha_earth_ruche_h512_h256_lr0.001	test_tessera_ruche_h512_h256_lr0.001	test_alise_ruche_h512_h256_lr0.001
Meadow	**0.9217**	**0.8859**	0.9216
Soft winter wheat	0.8793	0.8501	0.9155
Corn	0.9184	0.8572	0.9306
Winter barley	0.6718	0.7947	0.8354
Winter rapeseed	0.9124	0.8734	**0.9361**
Spring barley	0.5308	0.6404	0.7050
Sunflower	0.6303	0.7465	0.6896
Grapevine	0.8745	0.8561	0.8803
Beet	0.8953	0.8247	0.9289
Winter triticale	0.3222	0.4492	0.4334
Winter durum wheat	0.7250	0.7168	0.7979
Fruits, vegetables, flowers	0.4885	0.5821	0.5547
Potatoes	0.6144	0.5599	0.6988
Leguminous fodder	0.4339	0.5538	0.4998
Soybeans	0.8129	0.8238	0.8646
Orchard	0.7246	0.6713	0.6528
Mixed cereal	0.1472	0.4740	0.3414
Sorghum	0.2433	0.4725	0.4367



(a) MLP+ALISE-EMBEDDINGS



(b) MLP+AlphaEarth-EMBEDDINGS



(c) MLP+Tessera-EMBEDDINGS

Fig. 10 Confusion matrices on the test fold for foundation models.

Finally, we show in Fig. 11 and Fig. 12 the predicted semantic segmentation maps on a test patch. The scattered pixels on Tessera predictions shows that it work at a pixel level, while the smoother predictions of ALISE and Alpha Earth show that they work at the patch level. Note that in those prediction maps, the background and void classes are masked out (transparent pixels), which lead to clearly defined fields boundaries. While this is convenient for visualization, it is not the case for industry applications, where we often need to predict the background and void classes as well. Thus, it would be interesting to evaluate the performance of the models on those classes as well, which is left for future work. In the scarce data setting, we can see that the predictions are much noisier for U-TAE, which is consistent with the performance drop of U-TAE in the scarce data setting compared to the FM-EMBEDDINGS+MLP models.

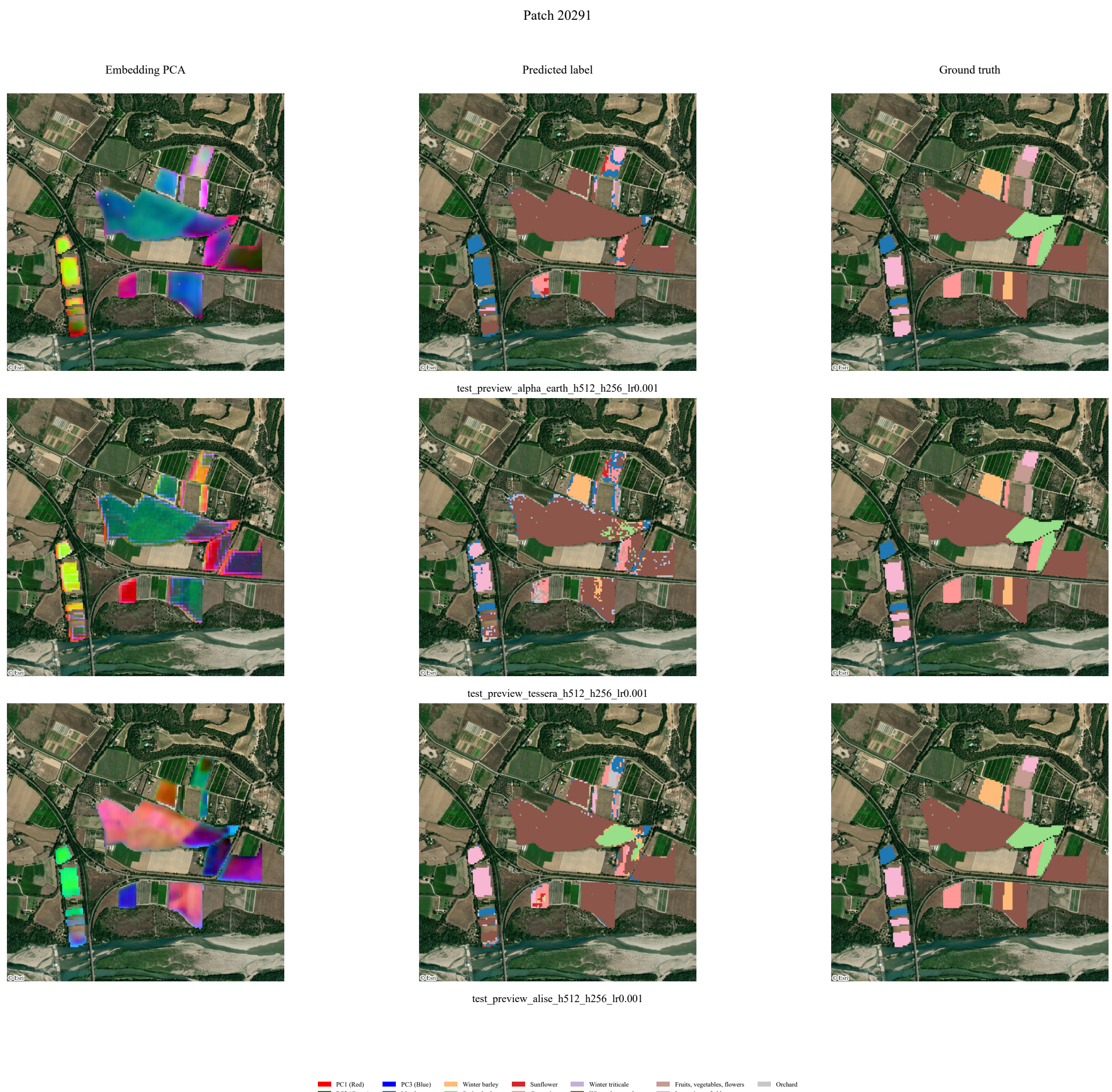


Fig. 11 Semantic segmentation map (full dataset). Row 1 : Alpha Earth, Row 2 : Tessera, Row 3 : ALISE. The first column shows the first three PCA components of the embedding, the second column shows the predicted segmentation map, the third column shows the ground truth segmentation map.

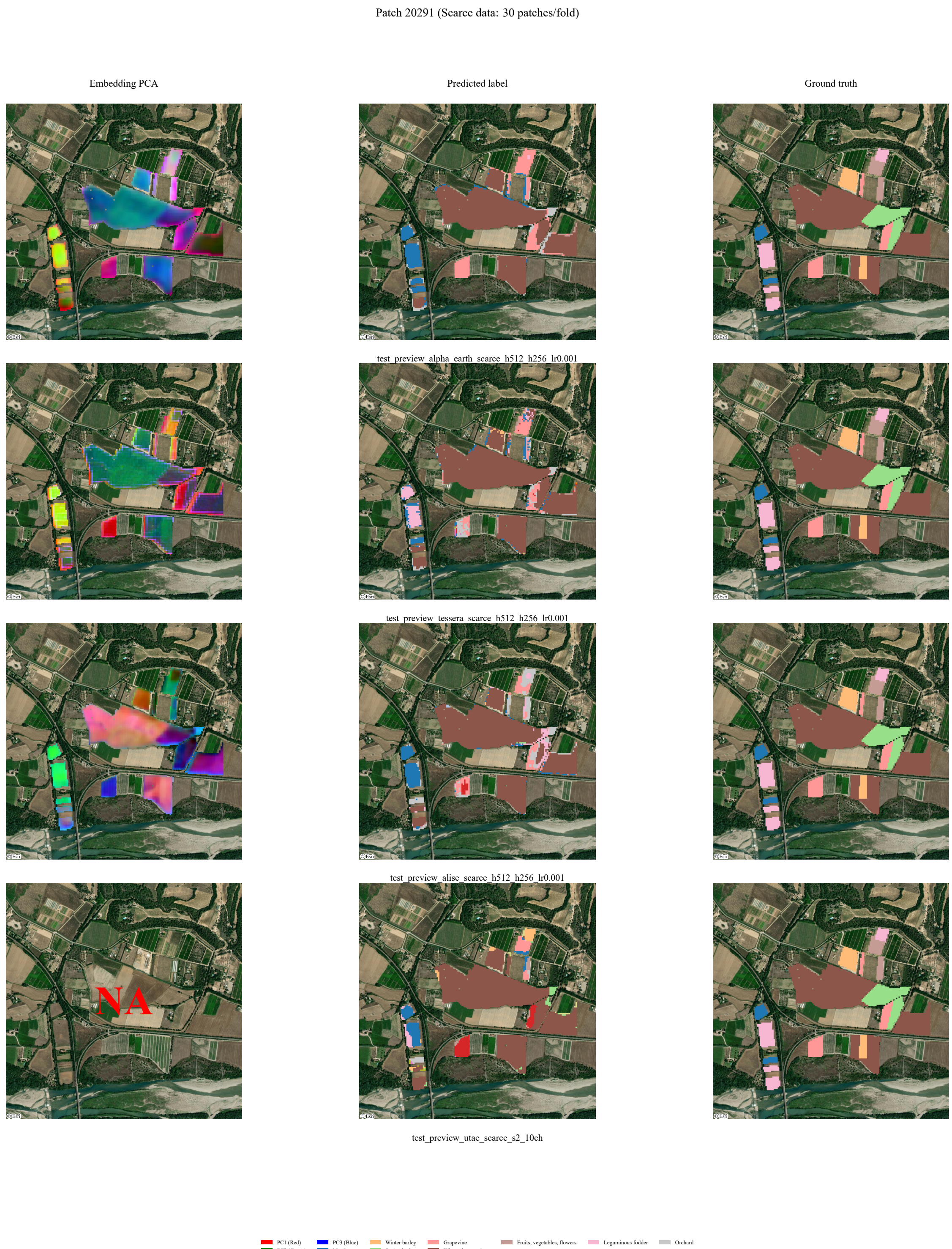


Fig. 12 Semantic segmentation map (scarce dataset). Row 1 : Alpha Earth, Row 2 : Tessera, Row 3 : ALISE, Row 4 : U-TAE. The first column shows the first three PCA components of the embedding, the second column shows the predicted segmentation map, the third column shows the ground truth segmentation map. Note that U-TAE embeddings are not available, as U-TAE is trained directly on the optical channels of the Sentinel-2 satellite.

4 Summary of contributions and future work

In this report, we propose a standard benchmarking framework to evaluate the performance of foundation models on the task of satellite image segmentation, reproducible and maintainable over time. We evaluate three state-of-the-art foundation models (TESSERA, Alpha Earth and ALISE) on the PASTIS dataset, and compare their performance to a supervised baseline (U-TAE). We show that while supervised methods achieve the best performance on the full training set, FM performs best in the scarce data setting.

Future work could include evaluating the performance of the models on other tasks, such as predicting the background and void classes. It would also be interesting to try other geographical regions, as the PASTIS dataset is limited to France, or even evaluate the transferability of the trained models to different regions on the same task. It would be interesting to include more foundation models in the benchmark as new models are released. We could further improve this benchmark by proving a website with an interactive interface to visualize the results and compare the different models, with a SDK to easily add new models to the benchmark, like https://timeseriesclassification.com/. The practicability of foundation models (e.g. easy training, whole Earth coverage, etc.) is good for prototyping applications, such as a crop monitoring application e.g, "what-is-this-crop.net"

References

[1] Iris Dumeur, Silvia Valero, and Jordi Inglada. *Paving the way toward foundation models for irregular and unaligned Satellite Image Time Series*. 2024. arXiv: 2407.08448 [cs.AI]. URL: https://arxiv.org/abs/2407.08448.

[2] Iris Dumeur, Silvia Valero, and Jordi Inglada. "Self-Supervised Spatio-Temporal Representation Learning of Satellite Image Time Series". In: *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 17 (2024), pp. 4350–4367. DOI: 10.1109/JSTARS.2024.3358066.

[3] Zhengpeng Feng et al. *TESSERA: Temporal Embeddings of Surface Spectra for Earth Representation and Analysis*. 2025. arXiv: 2506.20380 [cs.LG]. URL: https://arxiv.org/abs/2506.20380.

[4] Christopher F. Brown et al. *AlphaEarth Foundations: An embedding field model for accurate and efficient global mapping from sparse label data*. 2025. arXiv: 2507.22291 [cs.CV]. URL: https://arxiv.org/abs/2507.22291.

[5] Vivien Sainte Fare Garnot and Loic Landrieu. *Panoptic Segmentation of Satellite Image Time Series with Convolutional Temporal Attention Networks*. 2022. arXiv: 2107.07933 [cs.CV]. URL: https://arxiv.org/abs/2107.07933.

[6] Jason Ansel et al. "PyTorch 2: Faster Machine Learning Through Dynamic Python Bytecode Transformation and Graph Compilation". In: *29th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 2 (ASPLOS '24)*. ACM, Apr. 2024. DOI: 10.1145/3620665.3640366. URL: https://docs.pytorch.org/assets/pytorch2-2.pdf.

[7] William Falcon and The PyTorch Lightning team. *PyTorch Lightning*. Version 1.4. Mar. 2019. DOI: 10.5281/zenodo.3828935. URL: https://github.com/Lightning-AI/lightning.

[8] Lukas Biewald. *Experiment Tracking with Weights and Biases*. Software available from wandb.com. 2020. URL: https://www.wandb.com/.

[9] Vivien Sainte Fare Garnot et al. *Satellite Image Time Series Classification with Pixel-Set Encoders and Temporal Self-Attention*. 2019. arXiv: 1911.07757 [cs.CV]. URL: https://arxiv.org/abs/1911.07757.

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